

1. Consider a simple regression model $y_i = \beta_0 + \beta_1 x_i + u_i$. Let $\{(x_i, y_i): i = 1, \dots, n\}$ denote a random sample of size n from the population.

(a) Prove that given the data $\{(x_i, y_i): i = 1, \dots, n\}$, $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ minimize $\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.

Our objective function is:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n \hat{u}_i^2 = \min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0} = 2 \times \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \times (-1)$$

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1} = 2 \times \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \times (-x_i)$$

Let $\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0} = 0$ ---(1) and $\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1} = 0$ ---(2).

Solve (1) and (2),

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$
-----(1)

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \times (x_i) = 0$$
---(2)

We can get the answer $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Hessian matrix (for second partial derivative test) is

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_0} = 2 \sum_{i=1}^n (1), \quad \frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_1} = 2 \sum_{i=1}^n (x_i)$$

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1 \partial \hat{\beta}_0} = 2 \sum_{i=1}^n (x_i), \quad \frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1 \partial \hat{\beta}_1} = 2 \sum_{i=1}^n (x_i^2)$$

Because $\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_0} > 0$, $\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1 \partial \hat{\beta}_1} = 2 \sum_{i=1}^n (x_i^2) > 0$ and

$$2 \sum_{i=1}^n (1) \times 2 \sum_{i=1}^n (x_i^2) - 2 \sum_{i=1}^n (x_i) \times 2 \sum_{i=1}^n (x_i) > 0,$$

The answer indeed minimizes the objective function. For detail about this part, please check your calculus textbook or HE1004 note.

(b) Show that estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Plug $y_i = \beta_0 + \beta_1 x_i + u_i$ into the numerator,

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})\beta_0}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})\beta_1 x_i}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \frac{\beta_0 \sum_{i=1}^n (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} + \beta_1 \frac{\sum_{i=1}^n (x_i - \bar{x})x_i}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}\end{aligned}$$

We have simplified $\hat{\beta}_1$, and $\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$.

Take conditional expectation on both sides, the expectation of $\hat{\beta}_1$,
$$E[\hat{\beta}_1 | x] = \beta_1 + E \left[\frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} | x \right] = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) E[u_i | x]}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Under the zero conditional mean assumption $E[u_i | x] = 0$,

$$E[\hat{\beta}_1 | x] = \beta_1, \text{ i.e., } E[\hat{\beta}_1] = \beta_1.$$

Show that $\hat{\beta}_0$ is unbiased. We need to show $E(\hat{\beta}_0) = \beta_0$.

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

And we express $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = \beta_0 + \beta_1 \bar{x} + \bar{u}$,

$$\hat{\beta}_0 = \beta_0 + \beta_1 \bar{x} + \bar{u} - \hat{\beta}_1 \bar{x} = \beta_0 - (\hat{\beta}_1 - \beta_1) \bar{x} + \bar{u}.$$

Take conditional expectation on both sides,

$$E[\hat{\beta}_0|x] = E[\beta_0 - (\hat{\beta}_1 - \beta_1) \bar{x} + \bar{u}|x] = \beta_0 - E[(\hat{\beta}_1 - \beta_1) \bar{x}|x] + E[\bar{u}|x].$$

Since the second term

$$E[(\hat{\beta}_1 - \beta_1) \bar{x}|x] = E[(\hat{\beta}_1 - \beta_1)|x] \bar{x} = 0$$

and the third term

$$E[\bar{u}|x] = E\left[\frac{1}{n} \sum_{i=1}^n u_i \mid x\right] = \frac{1}{n} \sum_{i=1}^n E[u_i|x] = 0,$$

we obtain,

$$E[\hat{\beta}_0|x] = \beta_0, \text{ i.e., } E[\hat{\beta}_0] = \beta_0$$

(c) Derive $Var(\hat{\beta}_1|x)$.

Proof:

Based on (a), we know that $\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$.

$$Var(\hat{\beta}_1|x) = Var \left[\frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \mid x \right]$$

$$Var(\hat{\beta}_1|x) = Var \left[\frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \mid x \right]$$

$$\Rightarrow Var(\hat{\beta}_1|x) = \frac{Var[\sum_{i=1}^n (x_i - \bar{x}) u_i \mid x]}{[\sum_{i=1}^n (x_i - \bar{x})^2]^2} = \frac{\sum_{i=1}^n \{(x_i - \bar{x})^2 Var[u_i|x]\}}{[\sum_{i=1}^n (x_i - \bar{x})^2]^2}$$

$$\Rightarrow \text{Var}(\hat{\beta}_1|x) = \frac{\sum_{i=1}^n \{(x_i - \bar{x})^2 \text{Var}[u_i|x]\}}{\left[\sum_{i=1}^n (x_i - \bar{x})^2\right]^2}$$

With the homoskedasticity assumption $\text{Var}[u_i|x] = \sigma^2$, we obtain

$$\text{Var}(\hat{\beta}_1|x) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \sigma^2}{\left[\sum_{i=1}^n (x_i - \bar{x})^2\right]^2} = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

2. Explain what factors affect the magnitude of $Var(\hat{\beta}_1|x)$. In practice, how would you decrease the $Var(\hat{\beta}_1|x)$?

Recall:

$$Var(\hat{\beta}_1|x) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Clearly, when σ^2 decreases, x is more spread out, or the sample size increases, the variance drops.